TrustStroke: Trust-Aware Classification & Adaptive Decision Framework

Target: Lifetime Stroke History (CVDSTRK3) | CDC BRFSS Multi-Year Survey

SECTION 1: POPULATION DATA INGESTION, AVAILABILITY MODELING & SCENARIO RECALIBRATION

1. CDC BRFSS Ingestion & Design

- Sources: 2024 Source & 2025 Temporal Release
- Target: CVDSTRK3 (Self-reported stroke history)
- 15 Canonical Predictors (4 Categorical, 11 Numeric)
- Complex Survey Design: Weights, Strata & Clusters

2. Simulated Availability Scenarios

- Full (15 variables; No Demographics Only/No Variables)
- Demographics + Disease History (7 variables)
- Without Health State (12 variables; drops general/physical)
- Controlled missingness encoding (zero-masked & indicators)

3. Estimators & Scenario Recalibration

- Model Families: Boosted Trees, MLP, Logistic Regression
- 4 Training Controls: Global, Scenario, Mixed-Mask, Subset
- Platt Sigmoid Recalibration: $P_{cal} = \sigma(A_s \cdot \text{logit}(p) + B_s)$
- Resolves input representation mismatch without refitting

Patient Record x_obs

Scenario Calibrator Routing

Uncertainty Low (Skip Acquisition directly to Safety Gate)

SECTION 2: RUNTIME ADAPTIVE DECISION POLICY (Sequential Online Inference Loop)

Observed Input

Initial subset x_obs
Unobserved variables
concealed as NaN
(No feature leakage)

CAS Uncertainty Gate

Initial calibrated p & entropy
Check Acquisition Trigger:
H_total(p) ≥ τ_acq (0.30 bits)
or Disagreement ≥ τ_disagree

High Uncertainty

Feature Recommender

Rank missing candidates:
• Static Priority: CVDINFR4
• ConditionalDonors: max ΔH
(k=64 nearest complete donors)
Stop if ΔH < γ_min

Query c*

On-Demand Query

Strict Callback Provider:
v* = Query(c*)
Update x_obs ← x_obs ∪ {c*}
Steps k ← k + 1 (k ≤ K_max)

Safety Abstention Gate

Post-Acquisition Re-scoring
Check Abstention Criteria:
H_total(p) ≥ τ_abs (0.85 bits)
or Margin |p - θ| ≤ δ_margin

Update x_obs & Recalibrate (k < K_max)

Ambiguous / Boundary

Safe (H < τ_abs)

ACTION: ABSTAIN

y_hat = None | Emits Record

ACTION: CLASSIFY

y_hat = I(p ≥ 0.14) | Emits Record

SECTION 3: OFFLINE TRUST DIAGNOSTICS (Model Auditing; Excluded from Runtime Routing)

Decision Stability (DSI)

- Permutation Shapley attribution on fidelity
- Diagnoses Monte Carlo path noise (16 to 256)
- Evaluates retraining stability (seeds/bootstraps)

Confidence Attribution (CAS)

- $H_{total} = H_{within} + H_{between}$
- Decomposes ensemble entropy into member entropy & Jensen-gap disagreement
- Diagnostic audit tool for misclassification detection

Resource Adaptability (RAI)

- Brier headroom retention r_s above null prevalence baseline
- Evaluates multiple missingness patterns (MCAR & informative)
- Complements scenario-level proper scoring curves

** DSI, CAS decomposition, and RAI are offline diagnostics; they do not control online runtime routing.*

SECTION 4: DUAL INDEPENDENT EVALUATION REGIMES (Separately Evaluated Protocols)

Regime A: Forward Temporal Transport

- Core Conference Paper Evaluation
- 2024 Development → 2025 External Transport
- Complex Survey Weighting & Multi-Stage Clusters
- Stratified Taylor Linearization (Discloses Singletons)
- Compares FG vs. SC vs. MT vs. SR Controls

Regime B: Sequential Adaptive Benchmark

- Exploratory Dynamic Acquisition & Safety Study
- 2024 Locked Test Partition (I.I.D. Resampling)
- Paired Respondent-Level Bootstrap Differences
- Evaluates ConditionalDonors vs. Static Priority
- Documents Trade-offs of Safety Abstention

** Regime A (Survey-Weighted Temporal) and Regime B (I.I.D. Adaptive) are distinct protocols and not combined.*